

Pre-A Q3LOCK Fifth-History, Rank-Two Gap, and M2 Response-Boundary Checkpoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This single gate-level checkpoint synthesizes EXP-000813 and EXP-000814. It advances reusable result R-167 v2.2 and reusable result R-168 v1.3 without changing a claim card or tier. Both results are T0, additive, and `claim_bearing: false`. Their stable result identifiers are

1. [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#);
2. [PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT](#).

R-167 v2.2 closes the two previously named fifth-moment inputs and their actual-history composition in a deliberately split scope. The static Gibbs moment and the resulting two-orientation twentieth history moment are proved only for the registered fixed-beta finite periodic Q3 volumes with compact sources. The direct simultaneous-shear theorem is uniform for tested nearest-neighbor bond subsets of finite subgraphs and periodic quotients of $\mathbb{Z}^3$, or for a separately declared graph family with a uniform cubic polynomial-growth bound. Maximum degree six alone is insufficient. The composition gives the explicit hard-cutoff corridor but not the split limit or all-exhaustion common alpha. A separate exact rank-two projection fixture shows that an even/odd local kernel doublet and local edge gap do not automatically imply a volume-uniform lattice gap; it leaves a connected oscillator-elimination/QPS successor open.

R-168 v1.3 closes five exact mathematical children. A continuous pointwise real one-dimensional $U(1)$ representation is trivial and the raw real $H^2(\mathbb{T}^3; \mathbb{R})$ field space is contractible. A one-Q trial has an exact auxiliary optimized curvature and, on a fixed torus, a discrete fixed-amplitude central secant. A finite-regulator response tensor is shifted arbitrarily by a target-blind quadratic contact without changing the linear probe or zero-source data. An analytic response map transports a linear critical input to a positive integer exponent, not automatically exponent one. Finally, six adjacent relative-ratio controls give an explicit relative log-slope envelope; six absolute errors alone do not. None of these children constructs a compact action, physical mode, physical response, candidate, prediction, or prospective freeze.

The exact v1.0 combined source/PDF remains prior evidence only for R-167 v2.1 and R-168 v1.2. It is not current evidence for R-167 v2.2 or R-168 v1.3 and cannot satisfy this checkpoint. No per-lemma or intermediate PDF is issued. This v1.1 fragment is the one combined synthesis source for the present logical checkpoint; its sibling PDF may be issued only after proof, formal-authority, component, integrated, generated-surface, source-form, freshness, dual-extraction, render-review, and release gates pass.

2 Closed children, negatives, and surviving gates

The three R-167 children closed by EXP-000813 are exactly

1. PA-CP1-ST8-Q3LOCK-TRANSLATE-UNIFORM-LOCAL-FIFTH-GIBBS-MOMENT-AND-ELLIPTIC-EMBEDDING;
2. PA-CP1-ST8-Q3LOCK-SIMULTANEOUS-BOND-SHEAR-FIFTH-GRAPH-PROPAGATION;
3. PA-CP1-ST8-Q3LOCK-ACTUAL-TWO-ORIENTATION-TWENTIETH-HISTORY-MOMENT-AND-HARD-CUTOFF-CORRIDOR.

Its new scoped negative result is exactly

NG-2026-08-11-PRE-A-ST8-Q3LOCK-FULL-OSCILLATOR-LOCAL-PARITY-DOUBLET-EDGE-GAP-AUTOMATIC-VOLUME-UNIFORM-LATTICE-GAP.

The five R-168 children closed by EXP-000814 are exactly

1. PA-M2-CI8-V0-REAL-SCALAR-INTERNAL-U1-TRIVIALITY-AND-NO-INTRINSIC-WINDING;
2. PA-M2-CI8-ONE-Q-AUXILIARY-PHASON-CURVATURE-AND-FINITE-TORUS-SECANT;
3. PA-M2-CI8-HELICITY-TENSOR-CONTACT-SHIFT-NONIDENTIFIABILITY;
4. PA-M2-CI8-ANALYTIC-MAP-INTEGERS-EXPONENT-TRANSPORT;
5. PA-M2-CI8-SIX-STAGE-RELATIVE-LOG-SLOPE-ERROR-TRANSPORT.

Its four new scoped negative results are exactly

1. NG-2026-08-11-PRE-A-M2-V0-ONE-REAL-SCALAR-AUTOMATIC-INTERNAL-U1-WINDING-AND-HELICITY;
2. NG-2026-08-11-PRE-A-M2-ONE-Q-PHASON-AUTOMATIC-PHYSICAL-SUPERFLUID-DENSITY;
3. NG-2026-08-11-PRE-A-M2-POSITIVE-LOCAL-INVERTIBILITY-AUTOMATIC-UNIT-EXPONENT;
4. NG-2026-08-11-PRE-A-M2-SIX-ABSOLUTE-ERRORS-AUTOMATIC-LOG-SLOPE-CONTROL.

The new successor gates that remain open are exactly

1. PA-CP1-ST8-Q3LOCK-CONNECTED-RANK-TWO-OSCILLATOR-ELIMINATION-QPS-NORM-AND-CUTOFF-COMPATIBILITY;
2. PA-M2-SUCCESSOR-SUBSTANTIVE-COMPACT-ACTION-BACKGROUND-PROBE-AND-WINDING-LAW;
3. PA-M2-SUCCESSOR-ORDERED-STATE-PHYSICAL-MODE-AND-RESPONSE-LIMIT;
4. PA-M2-SUCCESSOR-SIX-TERM-CRITICAL-ESTIMAND-ERROR-BUDGET.

The broader common-alpha, infinite-dimensional rank-two, broken-sector GNS, physical-response, and Round-1 parents remain open, including PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA, PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS, PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY, PA-M2-CI8-PHYSICAL-RESPONSE-CHANNEL-AND-ERROR-BOUND, and PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE.

3 Ninth-order virial identity and the actual local fifth moment

Write the finite-volume polynomial Schrodinger operator and virial observable as

$$H = \frac{1}{2\chi} \sum_i p_i^2 + V(q), \quad F_i = \partial_i V, \quad A_i = \frac{1}{2}(q_i p_i^9 + p_i^9 q_i). \quad (3.1)$$

On the Schwartz core, the exact sign and coefficient identity is

$$\frac{i}{\hbar}[H, A_i] = \frac{p_i^{10}}{\chi} - \frac{1}{2} \sum_{j=0}^8 \left(q_i p_i^j F_i p_i^{8-j} + p_i^j F_i p_i^{8-j} q_i \right). \quad (3.2)$$

The proof does not use cyclicity of a Gibbs trace with an unbounded A_i . Every polynomially confining finite-volume eigenvector is Schwartz; (3.2) is first paired with each eigenvector, summed with finite Gibbs weights only below the spectral cutoff $P_N = \mathbf{1}_{\{H \leq N\}}$, and taken to $N \uparrow \infty$ only after positive uniform bounds permit monotone convergence.

With anisotropic orders $\text{ord } p = 1/2$ and $\text{ord } q = 1/4$, every force word has order at most five. Normal ordering and scalar Young inequalities give

$$\left| \frac{1}{2} \sum_{j=0}^8 \langle q_i p_i^j F_i p_i^{8-j} + p_i^j F_i p_i^{8-j} q_i \rangle \right| \leq 9\epsilon \langle p_i^{10} \rangle + C_\epsilon \langle 1 + Q_{\text{star}(i)}^{20} \rangle. \quad (3.3)$$

Choosing $9\epsilon < 1/\chi$ and using the registered fixed-beta periodic compact-source coordinate exponential estimate yields a volume-, site-, and translate-uniform tenth momentum moment. The compact shifted quartic onsite family $k_h \geq 1$ then obeys the parameter-uniform Shubin graph induction

$$\|k_h^{m/2} \psi\|^2 \asymp \sum_{|\beta|+|\alpha|/2 \leq m} \|q^\alpha p^\beta \psi\|^2, \quad 0 \leq m \leq 5, \quad (3.4)$$

in both directions. Consequently, in the registered periodic family,

$$m_5 = \sup_{\Lambda, h, x} \phi_{\Lambda, h}(k_{x, h}^5) < \infty, \quad d_5 = \sup_h \| |q|^{10} k_h^{-5/2} \| < \infty. \quad (3.5)$$

This discharges the named static input; it is not an arbitrary-boundary Gibbs theorem.

4 Direct cubic-growth bond shear and the fifth graph bound

For a tested edge e , set $f_x = e^{-\mu d(x, e)}$, with both endpoints assigned weight one, and $K_e = \sum_x f_x k_x$. For a tested nearest-neighbor subset F of a finite subgraph or periodic quotient of $\mathbb{Z}^3$, put

$$V_F = -c \sum_{xy \in F} q_x \cdot q_y, \quad B_F(\delta) = e^{-i\delta V_F/\hbar}, \quad Q_x^F = \sum_{y: xy \in F} q_y. \quad (4.1)$$

Direct conjugation terminates after the second term:

$$B_F(\delta)^* K_e B_F(\delta) = K_e + \delta R_1 + \delta^2 R_2, \quad (4.2)$$

where

$$R_1 = \frac{c}{\chi} \sum_x f_x p_x \cdot Q_x^F, \quad R_2 = \frac{c^2}{2\chi} \sum_x f_x |Q_x^F|^2. \quad (4.3)$$

The R_2 term contains every $q_y \cdot q_z$ cross term; no diagonal-only reduction is used. In an R_1 monomial $f_x p_x q_y$, paying $f_x^{1/2}$ to p_x and $f_y^{1/4}$ to q_y leaves

$$\frac{f_x}{f_x^{1/2} f_y^{1/4}} \leq e^{\mu/4} f_x^{1/4}. \quad (4.4)$$

In an R_2 cross monomial, paying $f_y^{1/4} f_z^{1/4}$ leaves

$$\frac{f_x}{f_y^{1/4} f_z^{1/4}} \leq e^{\mu/2} f_x^{1/2}. \quad (4.5)$$

The K_e^5 multinomial supplies these weights; normal-ordering commutators have lower anisotropic order and $K_e \geq 1$ fills the slack. Cubic growth gives

$$\sum_x f_x^{1/4} \leq 2 \left(\frac{1 + e^{-\mu/4}}{1 - e^{-\mu/4}} \right)^3. \quad (4.6)$$

If one fifth-power word has n_1 anchors of type R_1 and n_2 of type R_2 , then $n_1 + n_2 \leq 5$, while the unreduced neighbor-incidence count is

$$n_1 + 2n_2 \leq 10, \quad 6^{n_1+2n_2} \leq 6^{10}. \quad (4.7)$$

Every nonbaseline word contains at least one explicit δ . Hence, for $|\delta| \leq T$,

$$|\langle \psi, (B_F(\delta)^* K_e^5 B_F(\delta) - K_e^5) \psi \rangle| \leq C_5(T, \mu) |\delta| \langle \psi, K_e^5 \psi \rangle. \quad (4.8)$$

The factor $|\delta|$ is load-bearing for uniform split histories. Applying (4.8) at both signs yields onto invariance of $D(K_e^{5/2})$ and

$$\|K_e^{5/2} B_F(\delta) K_e^{-5/2}\| \leq e^{C_5(T, \mu) |\delta|/2}. \quad (4.9)$$

The proof is uniform for the stated cubic graph families and bond subsets, not for arbitrary graphs of maximum degree six. On a six-regular tree the sphere sizes are $6 \cdot 5^{r-1}$; at $e^{-\mu/4} = 1/2$ the weighted sphere terms are $3(5/2)^{r-1}$ and diverge. This hostile is why cubic polynomial growth appears explicitly in the theorem scope.

5 Actual periodic history moment and hard-cutoff corridor

Combining the periodic static input with the direct shear theorem and the retained endpoint embedding gives

$$\boxed{M_{20} \leq 2d_5^2 e^{C_5 T} S_\mu^5 m_5.} \quad (5.1)$$

For the hard restricted tail on the edge set of $[-R, R]^3$,

$$\sigma_+(W_{R,L}^2) + \sigma_-(W_{R,L}^2) \leq 2916c^2 M_{20} R^6 L^{-16}. \quad (5.2)$$

At $L = R^{2/5}$ this is

$$2916c^2 M_{20} R^{-2/5} \longrightarrow 0, \quad \log \frac{(CR^{4/5})^R}{R!} = -\frac{1}{5} R \log R + O(R). \quad (5.3)$$

This is the actual two-orientation history theorem only in the registered fixed-beta periodic compact-source scope. It does not prove $n \rightarrow \infty$, all-shape Cauchy convergence, all-exhaustion common alpha, the group, inverse or generator, or a KMS quotient.

6 Rank-two local edge does not force a global lattice gap

On each site take $\mathbb{C}^2$, $n = |1\rangle\langle 1|$, and

$$|\phi_{xy}^-\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}}, \quad h_{xy} = |\phi_{xy}^-\rangle\langle\phi_{xy}^-| + n_x n_y. \quad (6.1)$$

This is a rank-two projection with spectrum $\{0, 0, 1, 1\}$ and an even/odd kernel doublet. For every finite connected graph G ,

$$\ker \sum_{xy \in E(G)} h_{xy} = \text{span}\{|\text{vac}\rangle, |W_G\rangle\}, \quad (6.2)$$

while on the one-particle sector the Hamiltonian is $L_G/2$. On a periodic d -torus of side L ,

$$\text{gap}(H_G) \leq 1 - \cos(2\pi/L) \leq \frac{2\pi^2}{L^2} \rightarrow 0. \quad (6.3)$$

The infinite-onsite lift adds $K = z \sum_{r \geq 2} (r-1)|r\rangle\langle r|$ through $(K_x + K_y)/z$ on each edge. It preserves the local rank-two kernel and local gap one while the gapless one-particle band survives. This rejects only the automatic local-edge-to-global-gap inference. It is not a Q3 locality, coercivity, or gap no-go.

The surviving successor must construct a parity-equivariant, boundary/source-uniform quasi-local U_Λ , a uniform high-sector gap, cutoff convergence, a phasewise GNS intertwiner, and a connected interaction with

$$\sup_x \sum_{X \ni x} |X| e^{a \text{diam } X} \|\Phi_{\text{eff}}(X)\| < \epsilon_{\text{QPS}} J. \quad (6.4)$$

That successor remains open.

7 Real-scalar $U(1)$ and winding boundary

Let $\rho : U(1) \rightarrow GL(1, \mathbb{R}) = \mathbb{R}^*$ be continuous and pointwise linear. Its image is compact and connected and lies in $\mathbb{R}_{>0}$. The logarithm maps it to a compact additive subgroup of $\mathbb{R}$, hence to $\{0\}$. Thus ρ is trivial.

The raw configuration space $H^2(\mathbb{T}^3; \mathbb{R})$ is contracted by $C_s(\phi) = (1-s)\phi$. It has no intrinsic winding sectors as a raw real-field target. These statements do not exclude a spatial translation phase of a patterned state, an emergent complex or two-component amplitude, a defect-complement configuration space, or an externally supplied compact field. They therefore provide neither a winding law nor physical helicity.

8 One-Q auxiliary curvature and finite-torus secant

For $s_i \in \{-1, +1\}$, define

$$\phi(x) = A \cos((qs + a) \cdot x), \quad S(a) = \sum_i (2s_i q a_i + a_i^2)^2, \quad f(A, a) = \frac{A^2}{4} \{r + cS(a)\} + \frac{3gA^4}{32}. \quad (8.1)$$

On the branch $g > 0$ and $r + cS(a) < 0$,

$$A_*^2 = -\frac{4(r + cS(a))}{3g}, \quad f_{\min}(a) = -\frac{(r + cS(a))^2}{6g}, \quad (8.2)$$

and

$$\text{Hess}_a f_{\min}(0) = -\frac{8rcq^2}{3g}I_3. \quad (8.3)$$

Continuous a is an auxiliary Bloch, supercell, or thermodynamic variable; it is not available on one fixed periodic torus. There $h = 2\pi/L$, $q = mh$, and only integer multiples of h are allowed. At fixed amplitude A_0 the coordinate central secant is

$$\frac{f(A_0, he_i) + f(A_0, -he_i) - 2f(A_0, 0)}{h^2} = \frac{cA_0^2}{2}(4q^2 + h^2). \quad (8.4)$$

The continuum curvature, finite-step excess, and relative correction are $2cA_0^2q^2$, $cA_0^2h^2/2$, and $h^2/(4q^2) = 1/(4m^2)$. The only canonical upstream inputs are

$$r = -3, \quad c = 2, \quad q = 5, \quad g = 7, \quad h = 1, \quad m = q/h = 5. \quad (8.5)$$

They derive

$$A_*^2 = \frac{4}{7}, \quad 2cA_*^2q^2 = \frac{400}{7}, \quad \text{secant} = \frac{404}{7}, \quad \text{excess} = \frac{4}{7}, \quad \text{relative} = \frac{1}{100}. \quad (8.6)$$

The independent Laurent expansion gives coefficients $\{-3 : 1/8, -1 : 3/8, 1 : 3/8, 3 : 1/8\}$ and therefore $\cos^3 \theta = (3 \cos \theta + \cos 3\theta)/4$. A nonzero cubic term generates a third harmonic, so the one-Q family is a variational trial rather than an automatic exact Euler solution. Its auxiliary phason curvature is not an internal- $U(1)$ helicity modulus or physical superfluid density.

9 Tensor response and quadratic-contact nonidentifiability

For a finite-volume, finite-regulator norm- C^2 family

$$H(A) = H_0 - \sum_i A_i J_i + \frac{1}{2} \sum_{ij} A_i T_{ij} A_j, \quad T_{ij} = T_{ji}, \quad (9.1)$$

the positive finite-beta free-energy curvature is

$$\Upsilon_{ij}^{(\beta)} = \frac{1}{V} \left(\langle T_{ij} \rangle_\beta - \int_0^\beta \langle \delta J_i(-i\tau) \delta J_j \rangle_\beta d\tau \right). \quad (9.2)$$

For a simple isolated ground state with positive gap,

$$\Upsilon_{ij}^{(0)} = \frac{1}{V} \left(\langle 0 | T_{ij} | 0 \rangle - 2 \text{Re} \sum_{n>0} \frac{\langle 0 | J_i | n \rangle \langle n | J_j | 0 \rangle}{E_n - E_0} \right). \quad (9.3)$$

For any target-blind real symmetric matrix D , replacing T_{ij} by $T_{ij} + V D_{ij} I$ leaves H_0 , every J_i , and the zero-source state and spectrum fixed, while shifting Υ by exactly D . Thus the linear probe tensor does not identify a physical response before the quadratic-contact/background convention is supplied. These finite-regulator formulas are not a present Lane-Q compact action, winding law, ordered state, or physical response.

10 Analytic integer exponent and six-stage relative envelope

Assume

$$\kappa(\tau) = C\tau(1 + o(1)), \quad C > 0, \quad R(\kappa) = b_n \kappa^n + O(\kappa^{n+1}), \quad R(0) = 0, \quad b_n > 0, \quad n \geq 1. \quad (10.1)$$

Then

$$R(\kappa(\tau)) = b_n C^n \tau^n (1 + o(1)). \quad (10.2)$$

The transported exponent is the positive integer n . Exponent one requires $n = 1$; $R'(0) > 0$ with local regularity is a sufficient condition. Positive one-sided local invertibility alone is insufficient: x^2 and x^3 are the exact hostile leading orders two and three.

For six exact and approximate positive stages $R_0, \dots, R_6$ and $\hat{R}_0, \dots, \hat{R}_6$, with $\hat{R}_0 = R_0$, define adjacent ratios $g_j = R_j/R_{j-1}$ and $\hat{g}_j = \hat{R}_j/\hat{R}_{j-1}$. At both scales $s \in \{\tau, \lambda\tau\}$, suppose

$$g_j(s) \geq m_j(s) > 0, \quad |\hat{g}_j(s) - g_j(s)| \leq \epsilon_j(s), \quad \delta_j(s) = \frac{\epsilon_j(s)}{m_j(s)} < 1. \quad (10.3)$$

For $\lambda > 0$, $\lambda \neq 1$, telescoping gives

$$L = \prod_{j=1}^6 \frac{1 - \delta_j(\lambda\tau)}{1 + \delta_j(\tau)} \leq \frac{\hat{X}(\lambda\tau)/\hat{X}(\tau)}{X(\lambda\tau)/X(\tau)} \leq U = \prod_{j=1}^6 \frac{1 + \delta_j(\lambda\tau)}{1 - \delta_j(\tau)}, \quad (10.4)$$

where $X = R_6$ and $\hat{X} = \hat{R}_6$. Hence

$$|\hat{\nu}_\lambda(\tau) - \nu_\lambda(\tau)| \leq \frac{\max\{-\log L, \log U\}}{|\log \lambda|}. \quad (10.5)$$

Exponent transfer requires every relative δ_j at both scales to tend to zero. Six absolute-error bounds do not imply this: for $X(\tau) = \tau$ and $\hat{X}(\tau) = \tau + \epsilon$, the approximate dyadic log slope tends to zero while the exact one equals one. The verifier also checks the exact denominator spelling `abs(log(lambda))` and the explicit 4096 endpoint envelope used by the canonical six-stage fixture.

11 Adversarial suites and exact boundary

The new v1.3 theorem layer has exactly 27 hostile mutations. They cover a nontrivial real-line $U(1)$ promotion, intrinsic raw-field winding, spatial translation promoted to an internal symmetry, physical promotion of the auxiliary phason, Hessian/secant sign or fixed-torus errors, reoptimized amplitude substitution, deletion of the third harmonic, nonsymmetric or sign-flipped contact shifts, omission of the finite-beta contact or ground-gap hypothesis, x^2/x^3 unit-exponent promotion, six absolute errors promoted to relative control, $\delta = 1$, missing positive floors or vanishing-relative conditions, $\lambda = 1$, unbound initial stage, missing adjacent-ratio factorization, and candidate or parent promotion.

The earlier fail-closed schema suites are retained rather than replaced: 28 prospective-freeze hostile classes, 7 map-only repair hostiles, 11 successor-design hostiles, the hardened 57-case physical-response contract suite, and 48 deterministic malformed/fuzz cases in each validation engine. These counts validate only exact algebra, fail-closed syntax, local artifact binding, and scope. They do not prove semantic physical correctness or create a candidate, prediction, target, or freeze.

12 Devil's-advocate review

1. **DISMISSED – unbounded Gibbs cyclicity:** the virial proof works eigenvector by eigenvector under a finite spectral cutoff and takes a monotone limit only after positive bounds.
2. **VALID with mitigation – critical order:** the force term reaches order five, but (3.3) isolates $9\epsilon p_i^{10}$ and the registered coordinate twentieth moment; $9\epsilon < 1/\chi$ absorbs exactly the critical piece.
3. **DISMISSED – R_2 diagonalization:** all $q_y \cdot q_z$ cross terms are retained, leading to the incidence bound $n_1 + 2n_2 \leq 10$ and the safe 6^{10} tuple count.

4. **UPHELD – degree-only uniformity:** the six-regular-tree $3(5/2)^{r-1}$ hostile diverges. The direct shear theorem therefore requires cubic polynomial growth, and the actual Gibbs/history composition remains periodic-Q3 scoped.
5. **DISMISSED – step-count blowup:** every nonbaseline fifth-power word contains δ , so the budget exponentiates total absolute split time, not the number of Trotter rungs.
6. **UPHELD – local-to-global gap:** a local rank-two parity doublet and local edge gap do not supply a quasi-local connected elimination, QPS norm, or volume-uniform lattice gap.
7. **DISMISSED – real-line internal phase:** compactness and connectedness force every continuous $U(1) \rightarrow GL(1, \mathbb{R})$ representation to be trivial.
8. **UPHELD – patterned phases:** raw-field contractibility does not exclude a spatial phason, emergent complex order parameter, or defect-sector topology; those require separate physical structure.
9. **VALID with mitigation – finite torus:** continuous a belongs only to an auxiliary Bloch/supercell/thermodynamic derivative. The fixed torus uses reciprocal-lattice steps and a fixed-amplitude secant.
10. **VALID with mitigation – tensor sign/contact:** this note freezes $+V^{-1} \text{Hess } F$ and includes T_{ij} ; the invariant statement is only $\Upsilon \mapsto \Upsilon + D$.
11. **UPHELD – unit exponent:** an analytic map transports the first nonzero Taylor order n ; positivity or invertibility alone does not force $n = 1$.
12. **UPHELD – six-stage convergence:** exponent transport requires vanishing relative errors at both scales. Absolute errors alone do not control a logarithmic slope near a vanishing estimand.
13. **UPHELD – physical promotion:** no compact action, background probe, winding law, ordered-state quotient, response limit, six-term vanishing error budget, admitted map, or prospective freeze has been supplied.
14. **UPHELD – parent scope:** eight closed children do not close any common-alpha, connected rank-two/QPS, broken-sector GNS, physical-response, Round-1, C6, CP1, physical Sector A, or Pre-A parent.

Independent adversarial review is invited especially on the virial sign and finite-cutoff passage, the critical Shubin–Young absorption, the R_2 neighbor allocation and cubic-growth constant, the connected-token proof in the rank-two fixture, the Kubo/contact convention, the fixed-amplitude torus secant, the analytic leading-order hypothesis, and the adjacent-ratio factorization.

13 Reproduction and exact verification pins

The R-167 v2.2 primary, non-importing independent, and integrated scripts are

1. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py](#);
2. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py](#);
3. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py](#).

The R-168 v1.3 primary, non-importing independent, and integrated scripts are

1. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol.py`;
2. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_independent.py`;
3. `codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_verify.py`.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
    'codes\foundations\pre_a_cp1_st8_q3lock_' +
    'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
$r168 = (
    'codes\foundations\pre_a_round1_' +
    'prospective_holdout_freeze_protocol'
)
& $py -X utf8 ($r168 + '.py')
& $py -X utf8 ($r168 + '_independent.py')
& $py -X utf8 ($r168 + '_verify.py')
```

The final R-167 outputs are primary PASS 253/253, independent PASS 154/154, and integrated PASS 279/279. The final R-168 outputs are primary PASS 423/423, independent PASS 446/446, and integrated PASS 349/349.

The six verification-script raw SHA-256 pins are

```
R-167 primary:
d9d65080f84c0408200ba64c81449263cfd87095d8bdf1620211bc6fab6d1058
R-167 independent:
74dc4a8758d204587963c4e41e720902fd0b66931c35024f7784adaaa09d0b38
R-167 integrated:
5985f84cdb427d1fb3b3ab8de49e025c0ef3b0767e4462879eaa77e5907ba1bc
R-168 primary:
69a9486b060c711679314806b302af85652c6d8317fccebbba83578b5b2d397a9
R-168 independent:
6b100dd08e3daac385fc67fa5627f0c9f8c5d9ff8aa2a416d30018e72a033c26
R-168 integrated:
34af34a2bb45c50b68af0db88dfaf51004c3ab33d49c2c38464dd2fbed4f618e
```

The integrated totals are final only after the same source/PDF metadata is pinned identically in both manifests, the PDF is fresh relative to this source, pypdf and pdfplumber extract every declared page as nonempty and recover every required token, all rendered pages receive visual review, stored component and integrated JSON files equal fresh normalized invocations, formal authorities and generated surfaces are current, and the release check passes. A failed check invalidates the total rather than weakening the gate.

No PDF is built by this source-authoring step. No per-lemma or intermediate PDF belongs to R-167 v2.2 or R-168 v1.3. The eventual v1.1 PDF, if all gates pass, is the one current gate-level checkpoint; v1.0 and all earlier PDFs remain strictly historical evidence.

14 Exact no-overclaim and dependency boundary

The common-alpha parent is **OPEN**. The actual periodic-history theorem does not prove the split limit, all-shape Cauchy convergence, all-exhaustion common alpha, group/generator completion, or fixed-beta KMS quotient.

The rank-two parent is **OPEN**. The local-edge counterfixture proves an automatic-inference no-go; it neither refutes nor constructs the Q3-specific connected oscillator elimination, quasi-local effective interaction, two-phase QPS transfer, or cutoff compatibility.

The spectral parent is **OPEN**. A local edge gap and local parity doublet are not an oscillator-lattice phasewise GNS gap, observed temporal mass, or physical mass gap.

The M2 physical-response parent is **OPEN**. Real-scalar triviality, raw-field contractibility, auxiliary one-Q curvature, response nonidentifiability, analytic integer-order transport, and a relative six-stage error theorem supply no compact action, physical mode, response limit, or six-term critical estimand enclosure.

The prospective parent is **OPEN**. No real prospective freeze, external target commitment, admitted candidate prediction, public remote anchor, authenticated receipt, disclosure, score, or selection exists.

Nothing in this checkpoint proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, physical vacuum selection, C6, CP1, physical Sector A, or Pre-A closure. The active A5-SECTOR-A-SYNTHESIS remains conditional at its registered scope.

15 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT; PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT.
Reusable results:	R-167 v2.2; R-168 v1.3.
Explorations:	EXP-000813; EXP-000814.
Precise statement:	Actual fixed-beta periodic-Q3 local fifth moment, cubic-growth subset-shear fifth graph transport, two-orientation twentieth history moment and hard corridor; local-edge-to-global-gap no-go; real-line U1 triviality and raw-field contractibility; one-Q auxiliary curvature and torus secant; tensor-contact nonidentifiability; analytic integer exponent and six-stage relative log-slope transport.
Claim context:	C6-SPACETIME-SIGNATURE; claim_bearing false.
Closed children:	Three R-167 and five R-168 children, exactly the eight identifiers listed in Section 2.
Negative results:	One R-167 and four R-168 scoped negatives, exactly the five identifiers listed in Section 2.
Open successor gates:	Connected rank-two oscillator elimination/QPS norm; substantive compact-action/background probe/winding law; ordered-state physical mode and response limit; six-term critical-estimand relative-error budget.
Scope:	T0 additive, claim-nonbearing analytic/exact result at fixed-beta periodic Q3 history scope, cubic-growth subset-shear scope, exact counterfixture scope, auxiliary one-Q/finite-torus scope, finite-regulator response scope, and analytic/error-transport scope.
Dependencies:	R-167 v2.1; R-168 v1.2; EXP-000811; EXP-000812; registered periodic Gibbs and onsite graph inputs; retained local edge, finite-torus fingerprint, schema, and prospective-freeze authorities.
Evidence grade:	ANALYTIC + EXACT + primary + non-importing independent + integrated, conditional on the single final checkpoint lifecycle.
Reproduction command:	Run the six commands in Section 13 from repository

Expected output:	root, primary then independent then integrated. R-167: PASS 253/253, PASS 154/154, PASS 279/279. R-168: PASS 423/423, PASS 446/446, PASS 349/349.
Falsification gate:	Any virial sign/factor, finite-cutoff passage, Shubin order/absorption, R1/R2 coefficient or cross term, cubic-growth weight, incidence/tuple count, linear-in-step factor, M20/corridor exponent, rank- two spectrum/kernel/gap, U1/topology scope, phason Hessian/secant, contact sign, analytic order, relative-envelope, hostile count, hash, assertion count, provenance, freshness, extraction, rendering, or scope mismatch.
Tier before / after:	C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:	No all-exhaustion common alpha, connected rank-two Q3 oscillator elimination, oscillator-lattice GNS gap, physical compact action or response, candidate, real prospective freeze, regulator removal, continuum, physical-empty sign, C6, CP1, physical Sector A, or Pre-A closure.
Next required action:	Prove the split/all-exhaustion limit; construct and control the connected oscillator elimination in one QPS norm; derive a substantive physical M2 action, mode, response limit, and six-term vanishing relative error budget; then obtain and verify a real prospective freeze before disclosure.
PDF policy:	No per-lemma or intermediate PDF. Historical v1.0 and earlier PDFs are prior evidence only; issue one v1.1 checkpoint only after every proof, formal, component, integrated, PDF, freshness, extraction, visual, and release gate passes.